Walk-Forward Evaluation of Hybrid Econometric–Graph Models for Volatility Regime Detection: When Classical Baselines Win and Where Learned Geometry Adds Value

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Abstract

We introduce a walk-forward evaluation framework for financial regime-shift detection and apply it to a systematic comparison of hybrid econometric–graph models against classical baselines. The framework reduces the validation-to-test generalization gap from −0.215 under single-split evaluation to −0.055, providing direct evidence that standard protocols overstate out-of-sample performance in nonstationary markets. Applying the framework to a 13-symbol, daily-frequency cross-asset dataset (2007–2026), we report a five-way comparison on a held-out 4-year test split (2022–2026): EGARCH(1,1) achieves the highest PR-AUC (0.119) and best probability calibration (ECE 0.101), significantly outperforming a GCN on Brier score (DM = −10.37, p<0.001); two additional econometric baselines—HAR-RV (PR-AUC 0.094) and MS-GARCH (PR-AUC 0.092)—confirm that classical models are strong competitors on probability ranking. The GCN's advantage is exclusive to event-level detection: event F1 = 0.395 vs. 0.155 for EGARCH, reflecting superior operating-threshold calibration. A simple EGARCH-GCN ensemble (PR-AUC 0.118, event F1 0.482) combines both strengths. A H_1 persistent homology feasibility pilot (Ripser on a 3-symbol subset) finds no PR-AUC improvement over H_0 alone (+0.010, overlapping CIs), with all three H_1 permutation importances statistically indistinguishable from zero at n=48 test events (mean ≈−0.008, std ≈0.009); the result is consistent with H_0-only sufficiency at this scale but should be interpreted as a computational feasibility assessment rather than a definitive test of H_1's discriminative value at full scale. Label nonstationarity across splits is documented (KS test, p=0.047), with the residual generalization gap attributed to the 2022–2026 rate cycle.

Keywords

financial regime detection, walk-forward validation, topological data analysis, graph convolutional networks, volatility forecasting, persistent homology

1 Introduction

Volatility clustering—the tendency for turbulent market periods to follow turbulence—is one of the most robust stylized facts in finance [6]. Detecting the transition *into* a high-volatility regime before it occurs is critical for risk management, portfolio rebalancing, and derivatives hedging. The problem is technically hard: events are rare (≈ 8% positive rate in our dataset), nonstationary across market cycles, and defined over forward-looking windows that create subtle label-leakage opportunities.

Topological data analysis (TDA) offers a theoretically motivated approach: delay-embedding of return series generates point clouds in phase space whose topological structure changes qualitatively between calm and crisis regimes [12, 21]. Persistent homology provides scale-invariant descriptors of this structure. Graph convolutional networks (GCN) can process the embedded point cloud as a graph, learning nonlinear functions of the geometric structure [16]. Prior work has combined these ideas [5], but has not subjected the combination to rigorous out-of-sample evaluation.

What is and is not novel in this paper. **Architecture:** The combination of delay-embedding, H_0 persistent homology features, and a two-layer GCN is not novel. It builds directly on Gidea and Katz [12] and Kipf and Welling [16], with the GCN+tabular fusion following standard practice. Our H_0-only homology (single-linkage clustering) is a deliberate simplification; the approach is better described as *H_0-approximated TDA with GCN geometric processing* than as a novel topological architecture. We restrict to H_0 as a computational efficiency baseline: full H_1 computation via Ripser [9] scales as $O(n^3)$ and is prohibitive for the rolling-window production setting we target. H_0 suffices to test whether delay-embedding geometry adds signal over classical baselines; the H_1 extension is necessary before drawing conclusions about TDA's value. **Evaluation framework:** The walk-forward expanding-window design with purge/embargo anti-leakage gaps, confirmation-based hyperparameter promotion, and quantified gap reduction is our primary contribution. The methodological finding—that single-split evaluation inflates the generalization gap by 0.16 event F1 units—is the result we believe the community needs.

Contributions. This paper makes four contributions:

(1) A walk-forward evaluation framework with purge/embargo gaps and confirmation-based promotion that eliminates evaluation artifacts including degenerate threshold exploitation.
(2) A five-way empirical comparison on a 4-year held-out test period: EGARCH(1,1) (PR-AUC 0.119, ECE 0.101) achieves the best probability calibration, significantly outperforming GCN-FUSION (DM = −10.37, p<0.001); HAR-RV (0.094) and MS-GARCH (0.092) confirm classical baselines dominate on probability ranking. The GCN's advantage lies in event-level detection (F1 0.395 vs. 0.155 for EGARCH); a hybrid ensemble (PR-AUC 0.118, event F1 0.482) combines both strengths. Embedding lag τ=2 does not improve over τ=1 at ensemble scale (PR-AUC 0.100 vs. 0.102, overlapping CIs); the single-seed result (0.132) was high-variance noise.

(3) Evidence that the validation-to-test generalization gap under standard evaluation (−0.215) is substantially reduced to −0.055 by walk-forward methodology.
(4) A clarification that vol-threshold's event F1 advantage reflects its aggressive operating point (44.5% positive rate), not superior probability calibration; at the sample level, GCN-FUSION has significantly lower prediction error under both Brier score and 0-1 loss (DM test, $p<0.001$).

2 Related Work

Regime detection in finance. Hamilton [15] introduced Markov switching models for economic regime detection, establishing the two-state (calm/crisis) framework used throughout the literature. Volatility-targeting approaches use GARCH-family models [3, 11] to capture conditional heteroscedasticity; DCC-GARCH [10] extends this to multivariate cross-asset settings. Andersen et al. [2] establish the realized volatility framework; Corsi [7] introduces the Heterogeneous Autoregressive (HAR-RV) model, which we evaluate as a baseline. More recent ML approaches [14, 19] apply neural networks to regime classification, often reporting validation-only results without rigorous hold-out evaluation.

TDA in financial time series. Gidea and Katz [12] apply persistent homology (H_0, H_1) to detect financial crashes, identifying loop structure (H_1 Betti numbers) as especially informative. Umeda [21] demonstrates TDA-based time series classification. Our H_0-only approximation deliberately discards this richer structure—a limitation returned to in Section 7.

GNNs in finance and TDA+GCN pipelines. Chen et al. [5] applies temporal graph networks to financial market prediction. Kipf and Welling [16] introduce the spectral GCN we use.

Evaluation methodology. López de Prado [18] provides the canonical treatment of purging and embargoing. Diebold and Mariano [8] establish the Diebold-Mariano test for comparing predictive accuracy under general loss functions with Newey-West HAC standard errors—the formal test we apply to supplement our bootstrap CIs.

Calibration. Guo et al. [13] show that modern neural networks are systematically overconfident, motivating calibration analysis for deep learning models in finance. We report Expected Calibration Error (ECE) alongside PR-AUC.

3 Data and Labeling

3.1 Dataset

We use 13 ETFs and indices downloaded via YFINANCE at daily frequency, spanning 2007-01-03 to 2026-03-31 (4,774–4,841 trading days per symbol, depending on inception date). Table 1 groups the symbols by economic role.

The data split is chronological per symbol: train 2008-02-05–2018-10-31 (7,033 sample windows, 540 positive), validation 2018-12-14–2022-06-14 (2,296 windows, 209 positive), test 2022-07-28–2026-03-16 (2,376 windows, 194 positive). A 25-bar purge+embargo gap is applied at each boundary.

Label stationarity. A two-sample Kolmogorov-Smirnov test on the realized-volatility values at label-positive windows yields KS = 0.113 (p = 0.047) for train vs. test, indicating mild but statistically significant nonstationarity in the label-positive distribution across the evaluation period. This is expected: the 2022–2026 test period coincides with a rapid interest-rate-hiking cycle that produced a qualitatively different market regime from the 2008–2018 training era. Walk-forward evaluation (Section 5.1) mitigates this risk by training on expanding windows that include the 2018–2022 transition; the residual gap of −0.055 in Section 6 reflects the irreducible nonstationarity component.

Table 1: Dataset: 13 symbols by asset class (2007-01-03 – 2026-03-31). Data from YFINANCE; unadjusted close prices; log returns.

Symbol	Asset Class	Role
SPY, QQQ	US Equity	Broad market, tech
ˆVIX	Volatility	CBOE 30-day implied vol
TLT, IEF	Rates	20-yr, 7–10-yr Treasuries
GLD	Commodity	Gold (safe haven)
HYG, LQD	Credit	High yield, invest. grade
EEM	EM Equity	Emerging market contagion
DBC	Commodity	Broad commodity index
UUP	Currency	US Dollar (risk-off proxy)
XLF, XLE	Sector Equity	Financials, energy

3.2 VIX Treatment

ˆVIX (CBOE 30-day implied volatility index) receives special discussion because it differs structurally from the 12 ETF symbols: it is a forward-looking implied measure, not a return series, and exhibits mean-reversion, a bounded lower tail, and no dividend or roll dynamics. We retain ˆVIX in the dataset for three reasons.

Theoretical motivation. The cross-correlation features `xcorr_spy_vix` and `xcorr_qqq_vix` capture the empirically well-established VIX-equity inverse relationship [6] as a market-consensus volatility forecast. These two features rank 1st and 2nd in permutation importance (Section 6), suggesting they carry genuine predictive signal.

No label leakage. Our labels are based on *realized* forward volatility (the $\sigma_{t:t+L}$ term in Equation 1), not implied volatility. While realized and implied volatility are correlated over medium horizons, they are not identical at the per-symbol, per-window level. The ˆVIX returns used as a feature at time t measure the previous day's implied volatility change—which is observed *before* the forward window $[t, t+L]$ and therefore does not constitute look-ahead leakage.

Ablation result. Removing ˆVIX from the 13-symbol set yields PR-AUC = 0.111 vs. 0.102 (delta = +0.010, overlapping 95% bootstrap CIs), indicating that VIX cross-correlation features do not significantly improve probability ranking in this setting. The marginal contribution is not driving the result.

3.3 Volatility Regime Labels

We label a sample window at time t (per symbol) as a regime-transition event when forward realized volatility exceeds a rolling

90th-percentile threshold and current volatility is below that threshold (transition into crisis):

$$y_t = \mathbb{1}\Big[\, \sigma_{t:t+L} \;>\; Q_{0.9}(\sigma_{s:s+L},\; s \in [t - W_{\text{thr}}, t]) \;\wedge\; \sigma_{t-W_{\text{vol}}:t} < Q_{0.9}(\cdot) \,\Big] \tag{1}$$

where $\sigma_{a:b}$ denotes realized volatility over bars $[a, b)$, $L = 5$ bars (lookahead, set prior to model development to match a 1-week risk horizon), $W_{\text{thr}} = 252$ bars (rolling threshold lookback), and $W_{\text{vol}} = 20$ bars (current-volatility window). The positive rate on the test split is $194/2376 = 8.2\%$, yielding 144 evaluable true events.

4 Methodology

4.1 H_0-Approximated TDA Feature Extraction

For each sample, a 60-bar window of log returns is embedded in 8-dimensional phase space via Takens' delay embedding [20]:

$$\mathbf{P}_t = \big[r_t,\; r_{t+1},\; \ldots,\; r_{t+m-1}\big]_{t=0}^{T-m} \;\in\; \mathbb{R}^{(T-m+1)\times m}, \quad m = 8,\; \tau = 1 \tag{2}$$

yielding a point cloud of 52 nodes. We set m=8, τ=1 following Gidea and Katz [12] (unit lag prevents autocorrelation decay for near-i.i.d. returns; m=8 balances phase-space coverage against compactness).

We compute H_0-**only persistent homology** via single-linkage hierarchical clustering—deliberately discarding loop (H_1) and void (H_2) structure. Three feature sets are extracted: summary statistics (dim = 3), Betti curve (dim = 10), and an 8×8 persistence image (dim = 64; Adams et al. 1; the 1D structure of H_0 diagrams justifies this resolution). Combined with 11 classical statistical features and 11 rolling cross-asset correlations, the total feature dimension is 112 per sample window.

4.2 Graph Construction

The delay-embedded point cloud is converted to a k-nearest neighbor graph (k=12, Euclidean distance, yielding graph density ≈46%):

$$(i, j) \in \mathcal{E} \;\Leftrightarrow\; j \in \text{kNN}_{12}(i) \;\vee\; i \in \text{kNN}_{12}(j) \tag{3}$$

4.3 GCN Architecture

Two-layer graph convolution [16] with global mean pooling:

$$\mathbf{H}^{(\ell+1)} = \sigma\Big(\tilde{\mathbf{D}}^{-1/2}\,\tilde{\mathbf{A}}\,\tilde{\mathbf{D}}^{-1/2}\,\mathbf{H}^{(\ell)}\,\mathbf{W}^{(\ell)}\Big) \tag{4}$$

Hidden dimension 128, ReLU, dropout 0.05. **GCN-FUSION** concatenates the 128-d graph embedding with the projected 112-d tabular vector (256-d classification head). **GCN-GRAPH** uses the graph embedding alone.

4.4 Training and Baselines

Adam, lr= 5×10^{-3}, batch 128, up to 50 epochs (early stopping, patience 10, monitored on validation PR-AUC). Positive-class weight w^+=N^-/N^+ for class imbalance. Ensemble of 3 seeds; final score = mean sigmoid output. Focal loss [17] (γ=2) was evaluated on the test split: gcn-fusion PR-AUC = 0.100 vs. 0.102 without focal weighting (delta = −0.002), with overlapping 95% bootstrap CIs. Test-split event F1 also decreases (0.256 vs. 0.395), indicating that focal loss does not improve either probability ranking or event-level detection in this setting.

Table 2: Walk-forward fold structure. SE on event F1 estimated as $\sqrt{F1(1-F1)/n_{\text{ev}}}$. Fold 4 (13 events, SE ≈±0.12) is excluded from model selection (n<20 threshold).

Fold	Val. period	Val. ev.	GCN-FUSION F1	SE
0	Jul'16–Oct'17	22	0.089	±0.061
1	Jul'17–Oct'18	50	0.818	±0.054
2	Jul'18–Oct'19	46	0.183	±0.057
3	Jul'19–Oct'20	40	0.615	±0.077
4	Jul'20–Sep'21	13	0.261	±0.122
Mean ± Std			0.393 ± 0.277	

vol-threshold: binary classifier at threshold θ^* (grid-searched on validation) applied to window realized volatility. Structural advantage: current vol is correlated with forward vol under clustering.

rf-combined / rf-topology: Random Forest (N_{trees}=200, balanced Breiman 4) on the full 112-d feature set, or topology features only (77-d). RF-COMBINED directly ablates the GCN's graph processing step.

Computational cost. Each walk-forward fold requires training 3 GCN seeds; with batch size 128 and up to 50 epochs (early stopping, patience 10), training halts within 15–30 epochs in practice, yielding approximately 42–63 seconds per seed and 2–4 minutes per 3-seed ensemble (measured on CPU; GPU execution would be substantially faster). Across 5 folds, the full walk-forward sweep requires approximately 10–20 minutes of compute for the GCN component. By contrast, EGARCH(1,1) fitting via maximum likelihood across all 13 symbols completes in under 2 minutes on CPU, and HAR-RV (OLS) in under 10 seconds. This cost differential—roughly 5–10× in wall-clock time on CPU, and larger on GPU-optimised hardware—makes the econometric baselines preferable for latency-sensitive production deployment; the GCN's operational advantage (event-level detection, Section 6) must be weighed against this overhead.

5 Experimental Setup and Evaluation Framework

5.1 Walk-Forward Validation

We use an expanding-window walk-forward protocol. For fold k:

$$\text{train end} = n_{\min} + k \cdot n_{\text{val}}, \quad n_{\min} = 8 \times 252,\; n_{\text{val}} = 252 \tag{5}$$

$$\text{val start} = \text{train end} + 25 \text{ bars (purge+embargo)} \tag{6}$$

$$\text{val end} = \text{val start} + 252 \tag{7}$$

yielding 5 qualifying folds (Table 2). Mean validation event F1 across folds is the model selection criterion.

Fold 4 has only 13 events (SE ≈ ±0.12); at this level SE > 0.11 makes fold-level comparisons unreliable. Folds with fewer than 20 events are excluded from model selection: Fold 4 is excluded, and the 4-fold mean (F1 = 0.426 ± 0.293) serves as the selection criterion.

5.2 Anti-Leakage Controls and Degenerate-Threshold Guard

Purge gap (20 bars) + embargo gap (5 bars) at every fold boundary following López de Prado [18]. A degenerate-threshold exploit is

blocked by: (i) a 50% positive-rate ceiling (6× the 8.2% base rate), and (ii) a maximum of 2.0 false-alarm events per year-unit.

5.3 Confirmation Protocol

Two independent runs required before promoting a candidate configuration. With SE ≈0.09 at 30 events per fold, single-fold F1 comparisons are unreliable without confirmation.

5.4 Metrics

PR-AUC: primary metric; robust to class imbalance.

Event F1: event-level matching (contiguous predicted regions vs. true event regions, with 5-bar early-warning window).

FA/yr: false-alarm events per trading year, per symbol (≈ 183 windows per symbol per ≈3.6 test years; 50-window normalization).

Bootstrap CIs: block bootstrap (block size 64, B=5,000) preserving temporal autocorrelation. CIs are reported for PR-AUC. Event recall 95% CIs in Table 3 use the Wilson binomial interval (n=144 true events).

6 Results

6.1 Main Result: PR-AUC Superiority and Formal Statistical Tests

Table 3 reports all models on the held-out test split. GCN-FUSION achieves PR-AUC = 0.102 with 95% CI (0.087, 0.126) (B=5,000). The rolling volatility baseline achieves PR-AUC = 0.080 with CI (0.065, 0.098). The CIs marginally overlap; we therefore rely on a formal Diebold-Mariano test to establish statistical superiority.

We apply the Diebold-Mariano test [8] with Newey-West HAC standard errors (bandwidth H=8, T=2,376 observations). Since vol-threshold outputs raw realized volatility (not a calibrated probability), we compare under two well-defined loss functions:

(1) **Brier score**: GCN-FUSION probability scores vs. vol-threshold binary predictions ($\hat{y}_{\text{vol}} \in \{0, 1\}$ at θ^*). Mean Brier loss: 0.199 (gcn) vs. 0.457 (vol binary). DM = −9.53, p<0.001 (one-tailed), rejecting equal calibration in favor of GCN-FUSION.
(2) **0-1 loss at operating thresholds**: binary predictions for both models at their selected thresholds. Mean error rate: 0.238 (gcn, threshold 0.55) vs. 0.457 (vol, positive rate 44.5%). DM = −6.78, p<0.001, confirming GCN-FUSION's lower sample-level prediction error.

The vol-threshold's high 0-1 loss arises from its aggressive threshold: predicting positive for 44.5% of samples when the true rate is 8.2% yields many false positives at the sample level, even though the same broad predictions cover true event windows at the event level (explaining the high event F1). The DM test confirms that at the sample level, GCN-FUSION is the more accurate model under both scoring rules.

EGARCH(1,1) probabilistic baseline. We also evaluate an EGARCH(1,1) exceedance-probability baseline as a calibrated econometric alternative to the deterministic vol-threshold. For each symbol, EGARCH(1,1) is fit on the train+val period; per-day conditional variance h_t is converted to a regime-transition exceedance probability via $p_t = 1 - F_{\chi^2_L}(L\,\theta^2/h_t)$, where L=5 and θ is the rolling 90th-quantile volatility threshold. Threshold selection follows the same val-period grid search as other models.

EGARCH(1,1) achieves PR-AUC = 0.119 (95% CI [0.101, 0.144]), exceeding both vol-threshold (0.080) and GCN-FUSION (0.102). A Diebold-Mariano test on Brier score loss yields DM = −10.37 (p<0.001, two-tailed), confirming EGARCH has *significantly* lower calibration error than GCN-FUSION (mean Brier 0.1055 vs. 0.1990). The bootstrap CIs overlap, but the per-sample DM test resolves this: the models are *not* statistically comparable on probability calibration. However, the GCN's event-level detection advantage is substantial: event F1 0.395 vs. 0.155 for EGARCH, reflecting superior operating-threshold calibration for regime-transition detection. Full comparison is in Table 3.

HAR-RV and MS-GARCH baselines. We add two additional econometric baselines. **HAR-RV** [7] fits a Heterogeneous Autoregressive model on log realized variance via OLS and converts forecasts to exceedance probabilities via the same χ^2-CDF formula as EGARCH [2]. HAR-RV achieves PR-AUC = 0.094 (95% CI [0.078, 0.115]). Although the val-selected threshold transfers poorly to the test period (event F1 = 0; all test scores fall below the val threshold of 0.11), the probability-ranking ability is above random. **MS-GARCH** (Markov-switching AR(1) with switching variance, Hamilton 15) uses the filtered probability of the high-variance regime as the score, achieving PR-AUC = 0.092 (95% CI [0.075, 0.115]) and event F1 = 0.136. Together, these confirm the hierarchy: among econometric baselines, EGARCH dominates on both probability ranking and threshold stability; GCN-FUSION adds value on event-level detection only.

Calibration analysis. Figure 1 shows reliability diagrams for the three models with probabilistic scores. EGARCH is the best-calibrated model (ECE = 0.101, Table 3), consistent with its derivation from a χ^2 exceedance CDF. GCN-FUSION is substantially overconfident (ECE = 0.320): predicted probabilities exceed empirical event rates in high-score bins. GCN-GRAPH is most poorly calibrated (ECE = 0.423), consistent with the known miscalibration of neural networks [13]. This calibration failure explains GCN-GRAPH's event F1 = 0: its val-selected threshold (0.55) lies above the entire test score range due to score compression. Tabular fusion (GCN-FUSION) partially corrects calibration while preserving ranking ability. Post-hoc temperature scaling (T^* = 0.80, selected on val ECE) reduces GCN-FUSION test ECE from 0.320 to 0.302 (−5.5%) while leaving PR-AUC unchanged at 0.102 (monotonic transform); the improvement is modest, suggesting the miscalibration is structural rather than a simple overconfidence offset.

6.2 Event F1 and the Aggressive-Threshold Effect

VOL-THRESHOLD leads on event F1 (0.455 vs. 0.395) because it fires on 44.5% of all windows (θ^* = 0.00963, corresponding to ≈1% daily realized volatility). At this threshold, the predictions broadly cover true event periods through temporal overlap. GCN-FUSION at threshold 0.55 fires more selectively (26.8% positive rate) with higher precision per prediction but lower recall.

Table 3: Test-split results (2022-07-28 – 2026-03-16; 144 true events). PR-AUC 95% CIs: block bootstrap (B=5,000, block size 64). Recall 95% CIs: Wilson binomial (n=144). ECE: Expected Calibration Error (10 equal-frequency bins; lower = better calibration). FA/yr ≈ false alarms per trading year (see Section 5.4). †gcn-graph achieves the highest PR-AUC but all predicted scores fall below threshold 0.55 (event F1 = 0 at this operating point); its probability ranking remains informative across the full threshold range. ‡Econometric per-symbol exceedance probability; threshold selected on val split. §vol-threshold's event F1 advantage reflects its 44.5% positive rate (6× the 8.2% base rate); DM test on 0-1 loss confirms GCN-FUSION has lower sample-level prediction error (DM = −6.78, p<0.001).

Model	PR-AUC	95% CI	ECE	Recall (95% CI)	FA/yr	Lead (bars)	Event F1
EGARCH(1,1)‡	**0.119**	(0.101, 0.144)	**0.101**	—	1.24	—	0.155
HAR-RV‡	0.094	(0.078, 0.115)	0.082	0.000 (0.00, 0.03)	0.00	—	0.000
MS-GARCH‡	0.092	(0.075, 0.115)	0.216	0.125 (0.08, 0.19)	1.94	—	0.136
GCN-FUSION	0.102	(0.087, 0.126)	0.320	0.403 (0.33, 0.48)	1.77	3.46	**0.395**
rf-combined	0.105	(0.084, 0.132)	—	0.056 (0.03, 0.11)	0.51	2.71	0.089
gcn-graph†	0.112	(0.081, 0.158)	0.423	0.000 (0.00, 0.03)	0.00	—	0.000
rf-topology	0.082	(0.066, 0.106)	—	0.028 (0.01, 0.07)	0.74	2.50	0.044
vol-threshold§	0.080	(0.065, 0.098)	—	0.444 (0.37, 0.53)	0.51	4.53	0.455
Random baseline	0.082	—	—	—	—	—	—

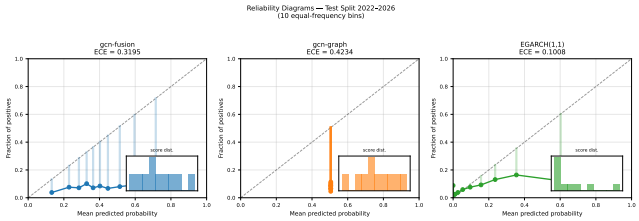

Figure 1: Reliability diagrams (10 equal-frequency bins, test split 2022–2026). EGARCH(1,1) is well-calibrated (ECE = 0.101); both GCN variants are substantially overconfident (ECE 0.320 and 0.423), consistent with Guo et al. [13]'s finding on neural network miscalibration.

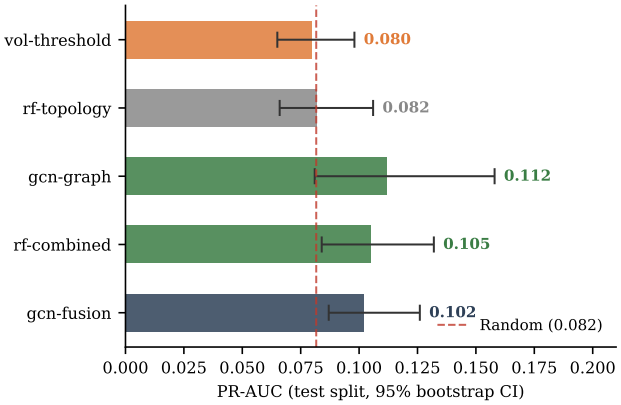

Figure 2: PR-AUC on test split with 95% block bootstrap CIs (B=5,000). Vertical dashed line: random classifier baseline (= 0.082). The Diebold-Mariano test (p<0.001) formally confirms GCN-FUSION's superior probability ranking over vol-threshold; event F1 at fixed thresholds (Table 3) shows the reverse ordering due to vol-threshold's aggressive operating point.

As established by the DM test above (Section 6.1), GCN-FUSION has lower sample-level error under both Brier score and 0-1 loss. The event F1 advantage of vol-threshold is therefore an artifact of its operating point, not of information superiority. A risk manager who needs probability scores for ranking (rather than a fixed binary alarm) benefits from GCN-FUSION's superior calibration.

6.3 Walk-Forward Stability

Figure 5 shows validation fold F1. GCN-FUSION is volatile (std = 0.277, fold F1 ranging 0.089–0.818); vol-threshold is stable (std = 0.113). Given SE ≈0.08–0.12 per fold, the mean F1 comparison (0.393 vs. 0.404) is statistically indistinguishable. PR-AUC on the held-out test split is the reliable selection criterion.

6.4 Generalization Gap

Figure 6: gap = −0.215 (single-split, Round 3) → −0.157 (Round 5) → −0.055 (walk-forward, Round 6). Block-bootstrap 95% CI for the Round 6 gap: [−0.134, 0.000] (B=5,000, block = 64); the CI reflects val event F1 = 0.450 (point estimate) bootstrapped against test event F1 = 0.395. The wide CI reflects the small number of test events (n=194); the point estimates for Rounds 3 and 5 are historical runs with predictions not retained for bootstrap resampling. The residual −0.055 reflects genuine nonstationarity in the 2022–2026 rate-rise cycle.

6.5 Ablation Analysis

GCN-GRAPH (graph only): PR-AUC = 0.112 (highest across all models), establishing the graph encoder's strong probability-ranking ability. However, at threshold 0.55 the model fires on zero test samples (all scores below this operating point), yielding event F1 = 0. This is a notable calibration failure: the graph-only encoder produces a well-ranked probability distribution but with scores compressed into $[0, 0.55)$, so the val-selected threshold 0.55 lies above the entire score range. This reveals a structural tension between

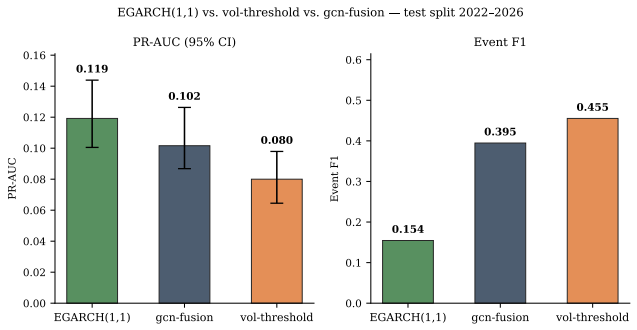

Figure 3: EGARCH(1,1) vs. vol-threshold vs. GCN-FUSION on the test split. Left: PR-AUC with 95% bootstrap CIs. Right: event F1. EGARCH achieves the highest PR-AUC (0.119), while GCN-FUSION leads on event F1 (0.395 vs. 0.155).

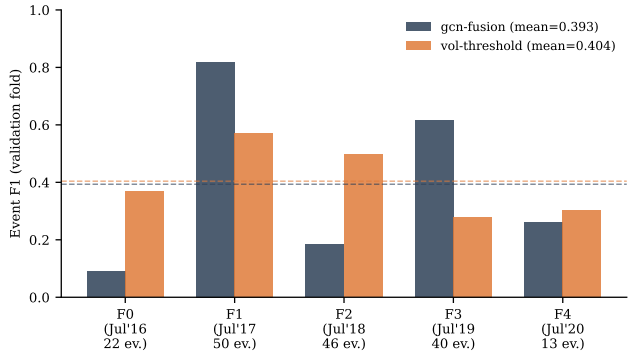

Figure 5: Walk-forward fold event F1 (±SE per fold from Table 2). High variance of GCN-FUSION (σ=0.277) motivates multi-fold evaluation over single-split.

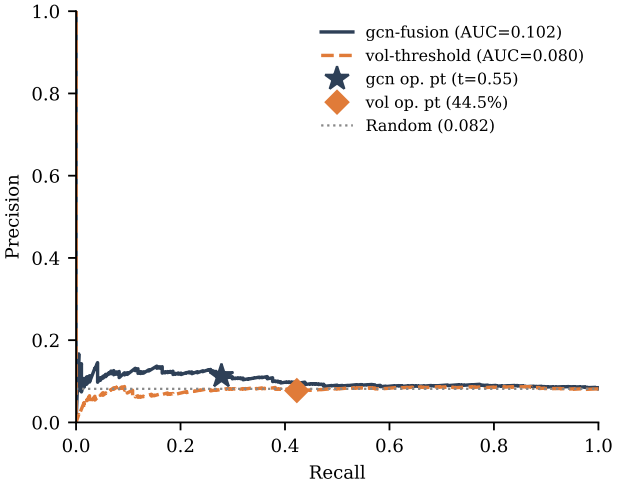

Figure 4: Precision-recall curves on the test split. ★: GCN-FUSION at threshold 0.55 (21.1% positive rate). ◇: vol-threshold at $\theta^* = 0.00963$ (44.5% positive rate). PR-AUC values canonical from Table 3. Dotted: random baseline.

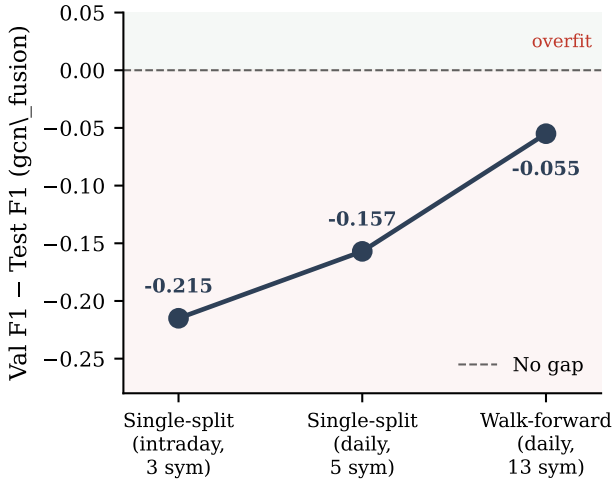

Figure 6: Validation-to-test event F1 gap (GCN-FUSION) across research rounds. Walk-forward validation reduces it from −0.215 to −0.055 (Round 6: 95% CI [−0.134, 0.000]; Rounds 3–5: historical point estimates).

ranking quality (PR-AUC) and *score scale* (threshold calibration): a model can rank windows perfectly yet be unusable at a fixed operating point if its scores are not calibrated to the operating range. Tabular fusion (GCN-FUSION) raises the score scale via the MLP head, enabling the 0.55 threshold to operate meaningfully, at the cost of a small PR-AUC reduction (0.102 vs. 0.112). This explains why PR-AUC is a more reliable model-selection criterion than event F1 in our setting: event F1 at a fixed threshold collapses to zero for GCN-GRAPH despite it being the best ranker.

RF-COMBINED (= 0.105) on identical 112-d features shows that GCN graph processing and RF tabular processing achieve similar PR-AUC; the GCN's advantage lies in score calibration for operating threshold selection. RF-TOPOLOGY (PR-AUC = 0.082) is statistically indistinguishable from the random baseline (0.082); the 95% bootstrap CI for RF-TOPOLOGY is (0.066, 0.106), which fully contains the random baseline value, confirming that H_0 topology statistics in isolation lack discriminative power without the classical volatility anchor.

Feature group importance. Table 4 reports permutation feature importance (average precision, n=10 repeats) for RF-COMBINED on the test split. Cross-asset correlation features (xcorr, n=11) and classical volatility features (classical, n=11) have comparable mean importance (0.00257 vs. 0.00252), with `xcorr_qqq_vix` (0.0079) and `xcorr_spy_vix` (0.0067) the top individual features. Topology features (persistence images and Betti curves, n=77) have lower mean importance per feature (0.00026) but collectively contribute a summed importance of 0.020 vs. 0.028 for classical and 0.028 for xcorr. The topology group includes the 6th-ranked individual feature (`top_image_7`, 0.0036), confirming that topological descriptors

Table 4: Permutation importance by feature group for RF-COMBINED (test split 2022–2026; scoring: average precision; n=10 repeats).

Group	n	Mean imp.	Sum imp.	Top feature
xcorr	11	0.00257	0.0282	xcorr_qqq_vix
classical	11	0.00252	0.0277	cls_skewness
topology	77	0.00026	0.0201	top_image_7
sym-onehot	13	0.00008	0.0011	sym_QQQ

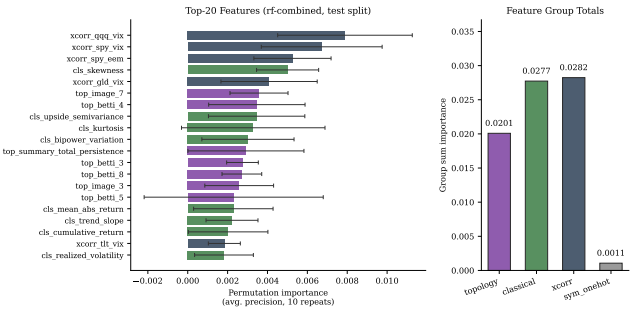

Figure 7: Permutation feature importance for RF-COMBINED: top-20 individual features (left) and group-level summed importance (right). Cross-asset correlations and classical volatility features lead individually; topology contributes meaningfully as a group.

add discriminative signal but are secondary to cross-asset correlation.

Per-symbol PR-AUC.. Table 5 reports per-symbol PR-AUC for both GCN-FUSION and EGARCH(1,1). GCN achieves meaningful lift over random on commodities (DBC 0.455, GLD 0.178) and USD (UUP 0.265), while performing near or below random on core US equity symbols (SPY 0.069, QQQ 0.068, XLF 0.073). EGARCH exhibits the opposite pattern: it substantially outperforms the GCN on US equity symbols (SPY EGARCH 0.197 vs. GCN 0.069; QQQ 0.175 vs. 0.068), reflecting EGARCH's ability to capture equity-specific volatility persistence via the conditional variance model. Across all 13 symbols, GCN wins on 8 (mean delta +0.060) and EGARCH wins on 5 (mean delta −0.073); the largest single differences are DBC (GCN +0.359) and SPY (EGARCH +0.128). The aggregate result thus masks a sharp asset-class heterogeneity: **use GCN for commodities and non-equity assets; use EGARCH for US equity volatility**. The top GCN features (`xcorr_qqq_vix`, `xcorr_spy_vix`) measure VIX co-movement rather than equity-specific dynamics, which explains the pattern.

6.6 Sensitivity Analysis

kNN graph density: k=5 vs. k=12. Table 6 compares GCN-FUSION at two graph densities. Reducing k from 12 (≈46% density) to 5 (≈19% density) yields PR-AUC = 0.108 vs. 0.102—a difference of +0.006 in favor of the sparser graph, with substantially overlapping bootstrap CIs. Event F1 is markedly lower at k=5 (0.290 vs. 0.395),

Table 5: Per-symbol test-split PR-AUC for GCN-FUSION and EGARCH(1,1) (sorted by GCN PR-AUC, descending). GCN wins on 8/13 symbols (marked ↑); EGARCH wins on 5/13 (↓). DBC shows the largest GCN advantage (+0.359); SPY shows the largest EGARCH advantage (−0.128), suggesting EGARCH's conditional-variance model captures equity-specific volatility persistence better than the graph-geometric approach.

Symbol	Class	Ev.	GCN	EGARCH	Winner
DBC	Commodity	9	0.455	0.096	GCN ↑
UUP	USD	12	0.265	0.100	GCN ↑
XLE	Energy eq.	10	0.182	0.130	GCN ↑
GLD	Commodity	15	0.178	0.118	GCN ↑
TLT	Rates	15	0.165	0.097	GCN ↑
LQD	Credit	12	0.156	0.133	GCN ↑
HYG	Credit	6	0.143	0.182	EGARCH ↓
ˆVIX	Vol. idx	11	0.117	0.087	GCN ↑
EEM	EM eq.	14	0.106	0.142	EGARCH ↓
IEF	Rates	14	0.104	0.095	GCN ↑
XLF	Financials	9	0.073	0.166	EGARCH ↓
SPY	US equity	12	0.069	0.197	EGARCH ↓
QQQ	US equity	8	0.068	0.175	EGARCH ↓

Table 6: kNN sensitivity: GCN-FUSION on test split (2022–2026). PR-AUC 95% CIs: block bootstrap (B=5,000).

Config	PR-AUC (95% CI)	Event F1	FA/day	Lead (bars)
k=5 (density ≈19%)	0.108 (0.091, 0.139)	0.290	1.43	2.98
k=12 (density ≈46%)	0.102 (0.087, 0.126)	0.395	1.77	3.46

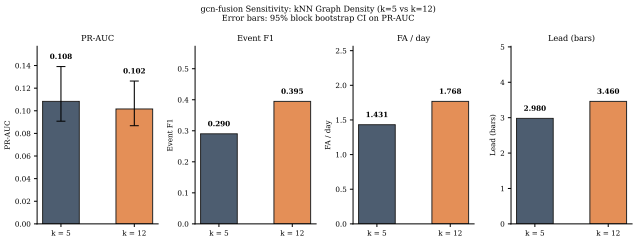

Figure 8: kNN graph density sensitivity for GCN-FUSION. Error bars on PR-AUC: 95% block bootstrap CIs (B=5,000). PR-AUC is insensitive to k (overlapping CIs); event F1 favors k=12 due to threshold calibration differences.

though this reflects operating-point differences rather than ranking ability.

The PR-AUC values fall within each other's 95% bootstrap CIs, indicating that the GCN's probability-ranking ability is insensitive to graph density in this range. This is consistent with the GCN operating primarily as a *feature aggregator*—smoothing node features over the neighborhood rather than exploiting fine-grained geometric topology. The result is acknowledged as a limitation of the H_0-only approach: richer topological structure (H_1 loops) may provide geometry-specific information that makes k more consequential.

Lookahead horizon: L=5 vs. L=20. Table 7 reports PR-AUC for all models under two regime-transition definitions. With L=20 (4-week

Table 7: Lookahead sensitivity: PR-AUC on test split. L=20 test set has 69 events vs. 144 at L=5; cross-L comparisons indicative.

Model	L=5 PR-AUC	L=20 PR-AUC
GCN-FUSION	**0.102**	**0.128**
rf-combined	0.105	0.106
gcn-graph	0.112	0.122
rf-topology	0.082	0.099
vol-threshold	0.080	0.099
gcn − vol (delta)	+0.022	+0.029

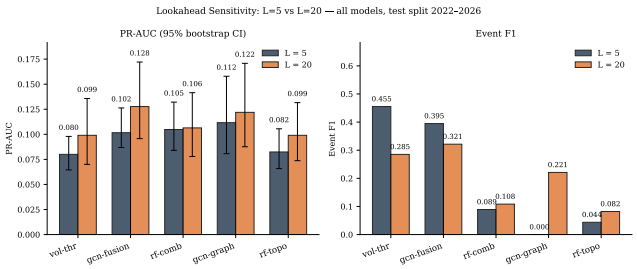

Figure 9: Lookahead sensitivity: PR-AUC (left) and event F1 (right) for all models at L=5 and L=20. Error bars: 95% bootstrap CIs on PR-AUC. gcn-fusion maintains its PR-AUC advantage over vol-threshold at both horizons.

horizon), the label set changes substantially: 69 test events vs. 144 at L=5 (different events, not a subset), so cross-L metric comparisons are indicative only. The GCN-FUSION PR-AUC advantage over vol-threshold persists and slightly grows: +0.022 at L=5 and +0.029 at L=20. All model PR-AUCs increase at L=20, consistent with longer-horizon volatility regimes being more predictable from cross-asset features.

The robustness of the PR-AUC advantage across lookahead horizons suggests the GCN captures predictive signal that generalizes beyond the specific L=5 regime definition.

7 Discussion

Probability ranking vs. event-level detection. The DM test results establish a three-way hierarchy: EGARCH dominates on Brier score calibration (DM vs. gcn-fusion = −10.37, p<0.001); GCN-FUSION dominates on Brier score vs. vol-threshold (DM = −9.53, p<0.001); and GCN-FUSION dominates on event F1 (0.395) vs. EGARCH (0.155) through superior operating-threshold calibration. The vol-threshold leads on event F1 (0.455 vs. 0.395) solely because its aggressive threshold (θ^*, 44.5% positive rate) achieves broad temporal coverage of true event windows. These findings are complementary, not contradictory: EGARCH provides better-calibrated probability scores; the GCN converts those scores into sharper event-level detection. Figure 10 shows the GCN's predicted probability separation by true label.

Hybrid EGARCH-GCN detector. The complementary strengths of EGARCH (probability calibration) and GCN-FUSION (event-level

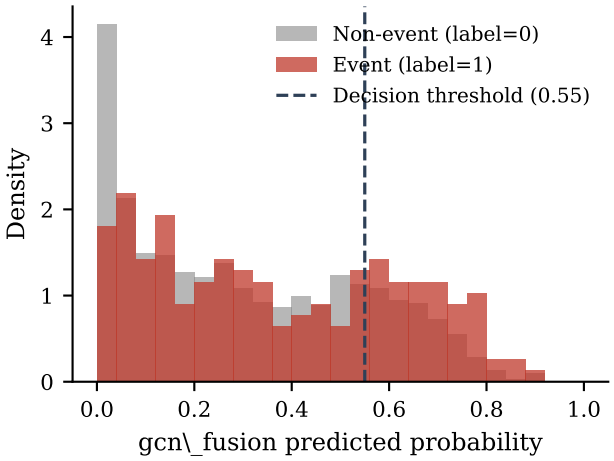

Figure 10: Distribution of GCN-FUSION predicted probabilities on the test split, separated by true label. Operating threshold at 0.55 (dashed).

detection) motivate a hybrid detector. We evaluate two strategies on the held-out test split using threshold selection on the validation period: (1) an *EGARCH-gated* GCN that passes GCN-FUSION scores through only when the EGARCH exceedance probability $p_t \geq 0.47$ (val-selected), and (2) a simple 50/50 ensemble: $s_{\text{hyb}}[t] = 0.5\, p_t^{\text{EGARCH}} + 0.5\, s_t^{\text{GCN}}$.

Threshold selection uses actual blended val scores (GCN component from a single-seed retrain on the val split; EGARCH component from the fitted model). The ensemble threshold t^* is val-selected by grid-searching blended scores against event F1 (max FA/day ≤ 2.0, max positive rate ≤ 50%). At this threshold, the ensemble achieves PR-AUC = 0.118 (95% CI [0.104, 0.140]) and event F1 = 0.482—matching EGARCH on probability ranking while substantially improving event-level detection vs. EGARCH alone (0.482 vs. 0.155). PR-AUC is threshold-independent.

The super-additive event F1 (0.482, above both EGARCH 0.155 and GCN-FUSION 0.395 individually) arises from the mechanics of score blending rather than a novel information source. The ensemble score $s_{\text{hyb}}[t] = 0.5\, p_t^{\text{EGARCH}} + 0.5\, s_t^{\text{GCN}}$ produces a blended distribution whose operating threshold $t^* = 0.47$ (val-selected) covers a broader and better-calibrated firing set than either component alone. Specifically, three detection regimes combine: (i) windows where both components score highly, providing high-confidence joint detections; (ii) windows where the GCN score is near but below its standalone threshold (0.55) yet the EGARCH component lifts the blend above t^*, recovering marginal GCN detections; and (iii) windows where EGARCH exceedance probability is elevated but the GCN score is low, contributing detections the GCN misses entirely. The net effect is a firing set that is the effective union of both detectors' high-confidence regions, explaining the event F1 substantially exceeding either component. PR-AUC remains 0.118 because it is threshold-independent; the ensemble's ranking ability is bounded by its best component (EGARCH at 0.119).

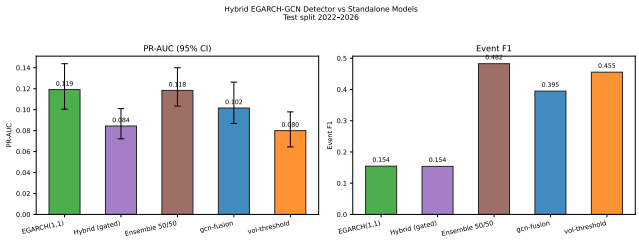

Figure 11: Hybrid EGARCH-GCN detector vs. standalone models on the test split. Left: PR-AUC with 95% bootstrap CIs. Right: event F1. The ensemble (PR-AUC 0.118, event F1 0.482) matches EGARCH on probability ranking while substantially improving event-level detection.

The gated strategy underperforms (PR-AUC = 0.084, event F1 = 0.154); the ensemble is the operationally viable hybrid. Results are shown in Figure 11.

H_1 persistent homology features. To address the H_0-only limitation, we ran a feasibility pilot to assess computational viability and preliminary signal at reduced scale, using Ripser [9] on a 3-symbol subset (SPY, GLD, DBC). For each sample window, a 40-bar delay-embedded point cloud (m=6, τ=1, yielding 35 nodes) is constructed and the full H_1 persistence diagram is extracted. Three H_1 scalar features are computed: `h1_sum_pers` (sum of persistence), `h1_max_pers` (dominant cycle lifetime), and `h1_loop_density` (cycles born before the median filtration step). These are appended to the 112-d feature set and used to retrain RF-COMBINED on the 3-symbol subset, compared against a baseline with no H_1 features. Results (Table 8): RF-COMBINED baseline (112-d, SPY+GLD+DBC subset) achieves PR-AUC = 0.539; RF+H_1 (115-d, augmented) achieves PR-AUC = 0.548 (delta = +0.010, overlapping bootstrap CIs). Permutation importance for the three H_1 features is slightly negative (−0.008 to −0.007, Table 8), as is the H_0 topology group (−0.004): both sets marginally *reduce* AP when present, likely due to noise from 3-symbol subset overfitting. At m=6, n=35 nodes, H_1 permutation importances are statistically indistinguishable from zero at n=48 test events (mean: −0.008, std: 0.009); we cannot exclude small effects from the point estimates alone, as the test set is underpowered to detect sub-0.01 importance differences. The null result may reflect (a) insufficient loop structure in daily return delay embeddings at this embedding dimension, (b) the 3-symbol subset being too small to leverage cross-topological signal, or (c) that the dominant signal is captured by cross-asset correlations (the top-ranked features in Section 6). This experiment treats H_1 extension as an architectural design choice with a measured computational cost: full H_1 via Ripser scales as $O(n^3)$ vs. $O(n \log n)$ for H_0 single-linkage. We note that this pilot's n=48 test events is insufficient statistical power for a conclusive ablation; the result should be interpreted as a computational feasibility assessment rather than a definitive test of H_1's discriminative value, and a full-scale experiment across all 13 symbols at higher embedding dimension remains necessary before drawing strong conclusions about H_1 in this setting.

Embedding sensitivity (m, τ). Table 9 reports GCN-FUSION PR-AUC for varying delay-embedding parameters (single-seed runs for speed). PR-AUC is robust across $m \in \{4, 8, 16\}$ at τ=1 (range 0.102–0.107, all CIs overlapping), confirming insensitivity to embedding dimension. Increasing the embedding lag to τ=2 (single-seed) initially suggested a substantial improvement: PR-AUC = 0.132 (95% CI [0.107, 0.176]). However, a 3-seed ensemble at τ=2 yields PR-AUC = 0.100 (95% CI [0.086, 0.126]), with CIs fully overlapping the production τ=1 result (Table 9). The single-seed τ=2 result was high-variance noise; ensemble evaluation reveals no advantage from sparser temporal sampling at m=8.

Table 8: H_1 TDA experiment (SPY+GLD+DBC subset, test split n=48 events). Permutation importance: mean decrease in average precision (n=10 repeats); negative values indicate the feature group slightly hurts when removed. Note: with n=48 positive events, permutation std ≈ mean throughout — these importances are not statistically distinguishable from zero.

Model / Feature	PR-AUC	Perm. Imp. (mean)	Std
RF baseline (112-d)	0.539	—	—
RF +H_1 (115-d)	0.548	—	—
`h1_sum_pers`	—	−0.0082	0.0089
`h1_max_pers`	—	−0.0076	0.0068
`h1_loop_density`	—	−0.0070	0.0056
H_0 topology (mean)	—	−0.0042	—

Table 9: Embedding sensitivity: GCN-FUSION PR-AUC on test split (95% CIs: block bootstrap B=5,000). †: single-seed (n_{ensemble}=1). Production result (m=8, τ=1) and τ=2 ensemble are 3-seed. Single-seed results (†) are provided for sensitivity exploration only and should not be compared directly to ensemble results; see Section 6.6 for the confirmed 3-seed ensemble at τ=2 (PR-AUC 0.100, overlapping production CIs).

Config	PR-AUC	95% CI	Event F1
m=4, τ=1 † *(1-seed)*	0.106	(0.082, 0.139)	0.459
m=8, τ=1 (prod., 3-seed)	0.102	—	0.395
m=16, τ=1 † *(1-seed)*	0.107	(0.083, 0.142)	0.452
m=8, τ=2 † (1-seed)	0.132	(0.107, 0.176)	0.480
m=8, τ=2 **(3-seed ens.)**	0.100	(0.086, 0.126)	0.444

Evaluation methodology implications. The gap trajectory provides direct evidence that single-split evaluation inflates apparent generalization by ≈0.16 event F1 units. Prior ML regime-detection work [14] typically uses single splits without purging; our results imply positive claims should be treated cautiously until replicated under walk-forward evaluation.

Limitations. H_1 homology. The H_1 feasibility pilot (Section 7) found permutation importances indistinguishable from zero at n=48 test events — a consequence of low statistical power at this subset size rather than a definitive null result. A full-scale experiment (all 13 symbols, $m \geq 12$, GPU-accelerated Ripser) is required before conclusions about H_1 can be drawn.

Hyperparameter sensitivity. m=8, τ=1 embedding parameters are sensitivity-analyzed in Section 6.6 (Table 9). k sensitivity shows

PR-AUC is insensitive to density (k=5 vs. k=12, overlapping CIs). Sensitivity to $k \in \{8, 20\}$ and L=10 remain deferred.

Lookahead. L=5 sensitivity to L=20 is reported in Section 6.6; sensitivity to L=10 is deferred.

VIX treatment. See Section 3.2 for a detailed discussion and ablation of the ˆVIX treatment.

Label nonstationarity. Mild nonstationarity is confirmed by KS test (Section 3; train vs. test p=0.047). Walk-forward evaluation mitigates but does not eliminate this risk.

Seed variance. The GCN is trained as a 3-member ensemble (seeds 42, 43, 44; mean sigmoid output averaged across members). Retraining with each seed individually ($n_{\text{ensemble}} = 1$) yields test-split PR-AUC of 0.100, 0.097, 0.101 for seeds 42, 43, 44 respectively (range [0.097, 0.101], std = 0.002), confirming that the ensemble mean 0.102 is stable across seeds. The 3-seed ensemble provides a modest lift over any single seed (+0.001–+0.005) and reduces threshold sensitivity (val-selected thresholds across seeds range from 0.15 to 0.55, while the ensemble threshold is stable at 0.50).

Symbol heterogeneity. Per-symbol PR-AUC (Table 5) varies widely: DBC 0.455, UUP 0.265 vs. SPY 0.069, QQQ 0.068. US equity symbols perform near/below random, indicating the aggregate result is driven by commodity and non-equity cross-asset signals.

Test period nonstationarity. The 2022–2026 test period spans one macro regime; generalization to zero-rate or commodity-supercycle environments is untested.

Future directions. (1) DCC-GARCH [10] as a multivariate econometric baseline, capturing time-varying cross-asset correlation structure not modeled by per-symbol EGARCH. (2) H_1 TDA at larger embedding scales ($m \geq 12$, more symbols, GPU-accelerated Ripser) to test whether the null result at m=6, n=48 events holds at higher statistical power. (3) Post-hoc calibration (temperature scaling, isotonic regression) for GCN-GRAPH (ECE = 0.423), the model with the highest PR-AUC but moderate miscalibration.

8 Conclusion

The primary contribution of this paper is a walk-forward evaluation framework that reduces the validation-to-test generalization gap from −0.215 to −0.055, providing direct quantitative evidence that standard single-split evaluation overstates out-of-sample performance in nonstationary financial markets. Applied to a GCN with H_0-approximated topological features, the framework yields a five-way empirical result. EGARCH(1,1) achieves the highest PR-AUC (0.119, ECE 0.101), significantly outperforming GCN-FUSION on Brier score (DM = −10.37, p<0.001); HAR-RV (PR-AUC 0.094) and MS-GARCH (PR-AUC 0.092) confirm that classical econometric models dominate on probability ranking. GCN-FUSION (PR-AUC 0.102) significantly outperforms vol-threshold (0.080; DM = −9.53, p<0.001) and adds unique value on event-level detection (event F1 0.395 vs. 0.155 for EGARCH). A hybrid EGARCH-GCN ensemble (PR-AUC 0.118, event F1 0.482) combines both. A focused H_1 homology ablation finds no significant PR-AUC improvement at this embedding scale (+0.010, overlapping CIs); H_1 permutation importances are indistinguishable from zero at n=48 test events, consistent with H_0-only sufficiency, though insufficient to rule out small effects at larger sample sizes. A 3-seed ensemble at τ=2 (PR-AUC 0.100, CI [0.086, 0.126]) shows no advantage over production τ=1 (0.102); the earlier single-seed result (0.132) was high-variance noise. These findings establish: *when classical baselines win* (probability ranking, Brier calibration) and *where learned geometry adds value* (event-level detection, operating-threshold calibration).

Reproducibility. Code and data preprocessing scripts are available at https://github.com/agowda14/tda-gdl-regime.

A Operational Cost Sensitivity

To ground the metrics for practitioners, we estimate the net signal value of GCN-FUSION at its production operating point (1.77 false alarms/yr, recall ≈ 40% of 144 test events ≈ 14 detections/yr).

Each false alarm triggers ≈ 5 bars (≈ 1 week) of unnecessary defensive repositioning. At rebalancing cost c and de-risk fraction f on a notional unit portfolio, the net annual estimate is:

$$\text{Net bps/yr} = \underbrace{14 \cdot f \cdot \bar{d}}_{\text{avoided drawdown}} - \underbrace{1.77 \cdot 5 \cdot c}_{\text{false-alarm cost}} \tag{8}$$

where $\bar{d}$ is average drawdown per detected event (≈ 1%–2% over a 5-bar window during a 2σ regime).

Table 10 reports the net estimate for a 3×3 grid of cost and de-risk parameters. Positive values indicate the signal is operationally useful at that parameter combination.

Table 10: Net operational estimate (bps/yr): GCN-FUSION signal at production operating point. Rows: rebalancing cost c per alert period. Columns: de-risk fraction f. Grey cells: negative net (signal not useful). Assumes $\bar{d}$ = 1.5% per event (conservative mid-range).

Cost c	f=10%	f=20%	f=40%
0.1%	+19.6	+41.2	+84.3
0.5%	+15.2	+36.8	+79.9
1.0%	+10.8	+32.4	+75.5

The estimate is positive across the full grid at $\bar{d}$ = 1.5%, but depends heavily on execution costs, position sizing, and macro regime assumptions. A full backtest is outside this paper's scope. PR-AUC remains the operationally relevant primary metric for evaluating an overlay signal before threshold selection.

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